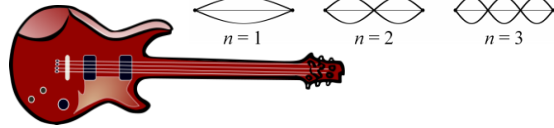


Fourier series example: guitar string



$$\Psi(x, t) = A^+ e^{i(\omega t - kx)} + A^- e^{i(\omega t + kx)}$$

$x = 0$ $x = L$

Boundary conditions:

$$0 = \Psi(x = 0, t) = A^+ e^{i(\omega t)} + A^- e^{i(\omega t)} \Rightarrow A^- = -A^+$$

$$0 = \Psi(x = L, t) = A^+ (e^{i(\omega t - kL)} - e^{i(\omega t + kL)}) = A^+ e^{i\omega t} (e^{-ikL} - e^{ikL}) = -A^+ e^{i\omega t} \sin(kL) = 0$$

$$\Rightarrow kL = n \cdot \pi, \quad n = 0, 1, 2, 3, \dots$$

The quantization of the k wavenumber also yields the quantization of the frequency due to the $\omega/k = v_{\text{wave}}$ condition. Accordingly, the wavefunction can include a standing wave with the base frequency, as well as higher harmonics:

$$\Psi(x, t) = \sum_n A_n \sin(k_n x) \cos(\omega_n t + \alpha_n), \quad k_n = n \frac{\pi}{L}, \quad \omega_n = n \frac{\pi v_{\text{wave}}}{L}, \quad n = 1, 2, 3, \dots$$

The time-dependent part remains periodic with $T = 2L/v_{\text{wave}}$ period time even after adding an arbitrary number of harmonics.

Reminder: different instruments played at the same musical frequency (fundamental frequency) have different tone colors because the different harmonics of the fundamental frequency. The tone color is attributed to the different amplitudes and phases of the higher harmonic components.

1

Fourier series

Statement: an arbitrary periodic function (for which $F(t+T)=F(t)$ for all t) can be expressed as the sum of harmonic functions including a DC component, a sine and cosine component with the base frequency, $f_1=1/T$ ($\omega_1=2\pi f_1$) and infinite sine and cosine components with the subharmonics of the base frequency, $f_n=n \cdot f_1$, $n=2, 3, 4, \dots$:

$$F(t) = \sum_{n=0}^{\infty} A_n^{\sin} \sin(\omega_n t) + A_n^{\cos} \cos(\omega_n t), \quad \omega_n = n \frac{2\pi}{T}, \quad n = 0, 1, 2, 3, \dots$$

Alternatively, one can use a complex notation. In this case the coefficient a_n is complex, the real part corresponds to the sine term, while the complex part corresponds to the cosine term. More precisely:

$$F(t) = \text{Re} \sum_{n=0}^{\infty} a_n \exp(i\omega_n t), \quad \omega_n = n \frac{2\pi}{T} = n \cdot 2\pi \cdot f_1, \quad n = 0, 1, 2, 3, \dots \quad a_n = A_n^{\cos} - iA_n^{\sin}$$

Calculating the complex coefficients (note, that for the $\omega_0=0$ DC term the sine component is anyhow zero, $\sin(\omega_0 t)=0$, i.e. a_0 can be taken to be real without the loss of generality!):

$$F(t) = \text{Re} \sum_{n=0}^{\infty} a_n \exp(i\omega_n t) = \frac{1}{2} \sum_{n=0}^{\infty} (a_n \exp(i\omega_n t) + a_n^* \exp(-i\omega_n t))$$

Let's multiply both sides by $\exp(-i\omega_m t)$ and integrate for an entire period time, T :

$$\int_{t_0}^{t_0+T} F(t) e^{-i\omega_m t} dt = \frac{1}{2} \sum_{n=0}^{\infty} a_n \int_{t_0}^{t_0+T} e^{i(\omega_n - \omega_m)t} dt + a_n^* \int_{t_0}^{t_0+T} e^{i(-\omega_n - \omega_m)t} dt = \begin{cases} T a_m / 2, & m > 0 \\ T a_0, & m = 0 \end{cases} \Rightarrow$$

$$a_m = \frac{2}{T} \int_{t_0}^{t_0+T} F(t) e^{-i\omega_m t} dt$$

$$a_0 = \frac{1}{T} \int_{t_0}^{t_0+T} F(t) dt$$

$\omega_m = m \frac{2\pi}{T}, \quad m = 0, 1, 2, 3, \dots$

This is how the Fourier coefficients are calculated

2

Examples of Fourier decomposition

Sawtooth signal

the first period is:

$$F(t, t = 0..T) = -\frac{A}{2} + \frac{A}{T} \cdot t$$

Fourier coefficients:

$$a_0 = \frac{1}{T} \int_0^T F(t) dt = 0$$

$$a_m = \frac{2}{T} \int_0^T F(t) e^{-i\omega_m t} dt = \frac{2A}{T^2} \int_0^T t e^{-i\omega_m t} dt = \frac{2A}{T^2} \left[\frac{e^{-i\omega_m t}}{\omega_m^2} (i\omega_m t + 1) \right]_0^T = \frac{2A}{T^2} \left[\frac{e^{-i2\pi m}}{\left(\frac{2\pi m}{T}\right)^2} (i2\pi m + 1) - \frac{1}{\left(\frac{2\pi m}{T}\right)^2} \right] = \frac{iA}{\pi m}$$

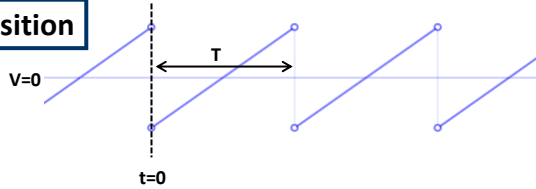
$$F(t) = -\frac{A}{\pi} \sum_{n=1}^{\infty} \frac{\sin(2\pi n t/T)}{n}$$

Symmetric triangle signal:

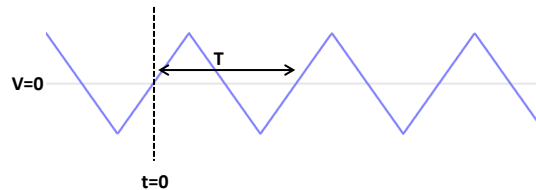
$$F(t) = \frac{8A}{\pi^2} \sum_{n=0}^{\infty} (-1)^n \cdot \frac{\sin(2\pi(2n+1)t/T)}{(2n+1)^2}$$

The even subharmonics are missing from symmetry reasons

Since there is no sudden change, it can be well approximated without high harmonics.



Experiment: sound and FFT of an asymmetric triangle signal



3

Fourier transformation

A function of period T can be described by Fourier decomposition with discrete components of circular frequencies spaced $\Delta\omega = 2\pi/T$ apart. For $T \rightarrow \infty$ $\Delta\omega \rightarrow 0$, i.e. any frequency ω can contribute to the function $F(t)$.

Definition: the Fourier transform of an $F(t)$

(not necessarily periodic) function:

(notation: upper case - time space, lower case - circular frequency space)

$$f(\omega) = \mathcal{F}(F(t)) = \int_{-\infty}^{\infty} F(t) e^{-i\omega t} dt$$

Example:

$$F(t) = e^{i\omega_0 t} \Rightarrow f(\omega) = \int_{-\infty}^{\infty} e^{i(\omega_0 - \omega)t} dt = \begin{cases} 0, & \text{if } \omega_0 \neq \omega \\ \infty, & \text{if } \omega_0 = \omega \end{cases} \stackrel{\text{def}}{=} 2\pi \delta(\omega_0 - \omega)$$

Definition: the inverse Fourier transform of the $f(\omega)$ function: $\mathcal{F}^{-1}(f(\omega)) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(\omega) e^{i\omega t} d\omega$

Fourier integral theorem (approximate): the inverse Fourier transform of the Fourier transform of a function returns the original function: $F(t) = \mathcal{F}^{-1}(\mathcal{F}(F(t)))$

$$F(t') = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega e^{i\omega t'} \int_{-\infty}^{\infty} dt F(t) e^{-i\omega t} = \frac{1}{2\pi} \int_{-\infty}^{\infty} dt F(t) \underbrace{\int_{-\infty}^{\infty} d\omega e^{i\omega(t'-t)}}_{2\pi \delta(t'-t)} = \int_{-\infty}^{\infty} dt F(t) \delta(t'-t) = F(t')$$

The above illustrates that the Dirac delta is a special function (distribution) for which:

$$\delta(x_0 - x) = \delta(x - x_0) = \begin{cases} 0, & \text{if } x_0 \neq x \\ \infty, & \text{if } x_0 = x \end{cases}, \text{ such that: } \int_{-\infty}^{\infty} \delta(x - x_0) dx = 1; \int_{-\infty}^{\infty} \delta(x - x_0) f(x) dx = f(x_0)$$

i.e. the Dirac delta is an infinitely narrow spike whose integral is 1.

More details in the *Mathematical methods in physics* course.

4

Supplement: proof of Fourier integral theorem (not very precise)

Let $F(t)$ be a continuous real function whose derivative is also continuous and whose absolute value is finite:

$$\int_{-\infty}^{\infty} |F(t)| dt < \infty$$

(It can be proven under stricter conditions, but that's all we're concerned with here.)

The inverse Fourier transform of the Fourier transform of $F(t)$: $\lim_{K \rightarrow \infty} \widetilde{F}_K(t')$, where:

$$\widetilde{F}_K(t') = \int_{-\infty}^{\infty} dt F(t) \underbrace{\int_{-K}^K d\omega \frac{e^{i\omega(t'-t)}}{2\pi}}_{\frac{\sin K(t'-t)}{\pi(t'-t)}} = \int_{-\infty}^{\infty} du F(t'-u) \frac{\sin Ku}{\pi u} = \int_{-\infty}^{\infty} du \underbrace{\frac{F(t'-u) - F(t')}{\pi u}}_{G(t',u)} \sin Ku + \frac{F(t')}{\pi} \int_{-\infty}^{\infty} du \frac{\sin Ku}{u}$$

This is a continuous, finite function of u , even at $u=0$ (here we get dF/du)

i.e.:

$$\lim_{K \rightarrow \infty} \widetilde{F}_K(t') = \lim_{K \rightarrow \infty} \underbrace{\int_{-\infty}^{\infty} du G(t',u) \sin Ku}_0 + F(t')$$

So the original function was indeed returned in the back-and-forth transformation!

5

Problems of real-world Fourier-analysis: finite measurement time

In real-world spectrum analysis we face two fundamental problems compared to mathematical Fourier analysis:

- (i) By nature, we can only measure for **finite time** instead of the infinite Fourier integral
- (ii) Instead of continuously measuring the signal, we can only **sample the data at discrete time intervals**.

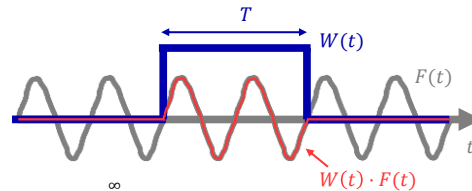
Let us investigate the consequences of (i) and (ii) separately.

First, let's assume that we sample continuously, but we measure for a finite T time.

The mathematical Fourier transformation:

$$f(\omega) = \int_{-\infty}^{\infty} dt F(t) e^{-i\omega t}$$

The finite measurement time is mathematically modeled by a **window function**, which is constant during the T time of the measurement, and zero otherwise ("rectangular window"). The Fourier transformation is calculated for the product of the $F(t)$ function and the $W(t)$ window function:



$$f_W(\omega) = \int_{-\infty}^{\infty} dt W(t) F(t) e^{-i\omega t}$$

Statement: the $f_W(\omega)$ Fourier transform of the product function ($W(t) * F(t)$) is practically the convolution of the $f(\omega)$ and $w(\omega)$ Fourier transforms of $F(t)$ and $W(t)$:

$$f_W(\omega) = \int_{-\infty}^{\infty} f(\omega') \cdot w(\omega - \omega') \frac{d\omega'}{2\pi}$$

Proof with Dirac-delta function:

$$\begin{aligned} f_W(\omega) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} dt dt' F(t) e^{-i\omega' t} \cdot W(t') e^{-i(\omega - \omega') t'} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dt dt' F(t) W(t') e^{-i\omega t'} \delta(t - t') = \\ &= \int_{-\infty}^{\infty} dt W(t) F(t) e^{-i\omega t} \end{aligned}$$

From the ω' integral

6

Finite time continuous Fourier transformation of a harmonic function

$$F(t) = Ae^{i\omega_0 t}, \quad W(t) = \begin{cases} 1/T, & \text{if } |t| < T/2 \\ 0 & \text{otherwise} \end{cases}$$

The Fourier transform of $F(t)$ is a Dirac-delta:

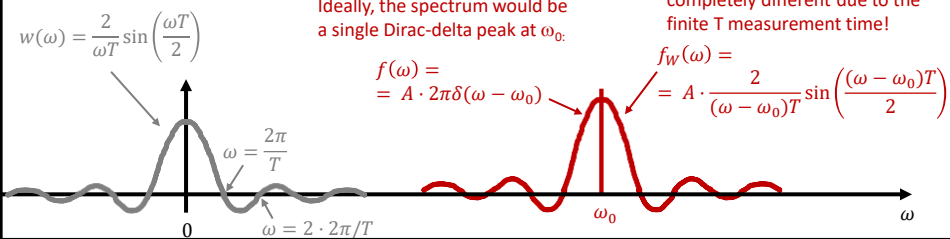
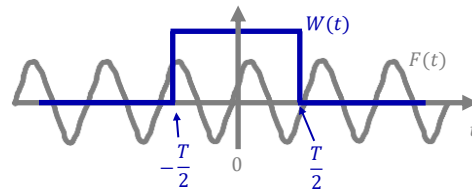
$$f(\omega) = A \cdot 2\pi\delta(\omega - \omega_0)$$

The Fourier transform of $W(t)$ is a $\sin(x)/x$ function:

$$w(\omega) = \int_{-\infty}^{\infty} dt W(t) e^{-i\omega t} = \frac{1}{T} \int_{-T/2}^{T/2} dt e^{-i\omega t} = \frac{2}{\omega T} \sin\left(\frac{\omega T}{2}\right)$$

The convolution of the two is the $\sin(x)/x$ function offset such that the central peak is shifted to ω_0 :

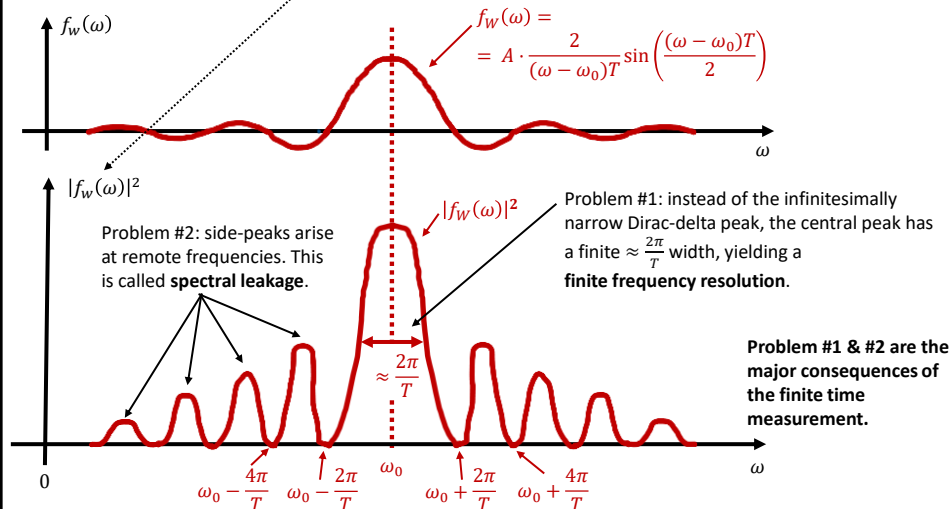
$$f_W(\omega) = \int_{-\infty}^{\infty} A \cdot 2\pi\delta(\omega' - \omega_0) \cdot w(\omega - \omega') \frac{d\omega'}{2\pi} = A \cdot w(\omega - \omega_0)$$



7

Finite frequency-resolution and spectral leakage

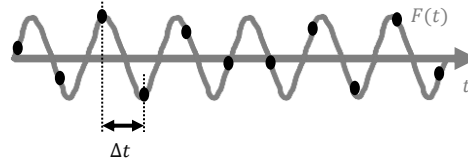
Further discussion of the previous slide: With a symmetric window function around the $t=0$ axis $w(\omega)$ is real, and thus $f_w(\omega)$ is also real. But at a general position of the window function a complex $f_w(\omega)$ is obtained. It is better to investigate $|f_w(\omega)|^2$, i.e. the "power" of the spectrum, which is independent of the position of the window function.



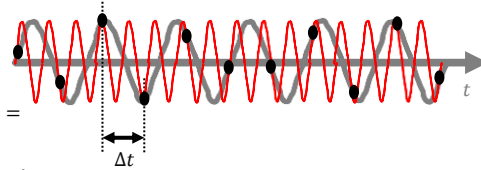
8

What, if we sample at finite discrete intervals, but for infinite time?

According to the **Nyquist-Shannon sampling theorem**: if a signal contains no components with frequencies higher than $f_{\max}$, then the signal can be perfectly described by its values sampled at intervals of $\Delta t = 1/(2f_{\max})$. Here $f_{\max} = 1/(2\Delta t)$ is called Nyquist frequency, or $\omega_{\max} = 2\pi f_{\max}$ is the Nyquist circular frequency.

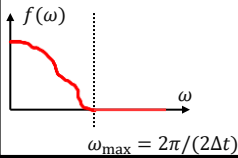


Aliasing: we sample a $F(t) = \sin(\omega t)$ signal at Δt intervals, $F_n = \sin(\omega n \Delta t)$. Problem: the same sample points are obtained for a much higher frequency signal, with $\omega' = \omega + 2k \cdot \omega_{\max}$, where k is an integer:



$$F_n = \sin((\omega + 2k \cdot \omega_{\max})n\Delta t) = \sin(\omega n \Delta t + kn2\pi) = \sin(\omega n \Delta t)$$

If our signal contains high frequency components with $\omega' > \omega_{\max}$, we cannot decide whether this high frequency signal, or the above $\omega < \omega_{\max}$ lower frequency signal is sampled.



If we have a band-limited signal, which does not contain any component at $\omega' > \omega_{\max}$, the finite Δt sampling is sufficient for the complete reconstruction of the full continuous spectrum, and the complete reconstruction of the temporal signal, once we collect infinite samples. Therefore, the discrete sampling with $N = \infty$ points yields a bandwidth limitation, i.e. the $f_{\max}$ Nyquist frequency limits our analysis, but the frequency resolution is not limited. The latter is related to the finite time of the measurement, and not the discrete sampling.

9

Discrete Fourier transformation (DFT)

The Fourier integral with a time window of length T can be replaced by a discrete sum, where $F(n\Delta t)$ is the n^{th} datapoint of the $F(t)$ signal, which is sampled at Δt intervals, and altogether N datapoints are collected during the time of the measurement (T):

$$f_W(\omega) = \int_{-\infty}^{\infty} dt W(t) F(t) e^{-i\omega t} \Rightarrow f_W(\omega) \approx \sum_{n=0}^{N-1} W(n\Delta t) F(n\Delta t) e^{-i\omega n \Delta t} \cdot \Delta t, \quad T = N\Delta t$$

N is taken as an even number! (We'll come back to that later.) Note, that the DFT includes both the finite measurement time and the discrete sampling.

If the samples are taken at discrete time intervals, there is no reason to continuously evaluate the spectrum in the circular frequency domain ($f_W(\omega)$). Instead, we should evaluate the spectrum at discrete frequencies starting at $\omega = 0$ and using circular frequency steps of our approximate frequency resolution, $\Delta\omega \approx \frac{2\pi}{T}$:

$$f_W(\omega_k = k \cdot \Delta\omega) \approx \sum_{n=0}^{N-1} W(n\Delta t) F(n\Delta t) \frac{e^{-ik\Delta\omega n \Delta t}}{e^{-ikn2\pi/N}} \cdot \Delta t, \quad \text{Suppose we evaluate as many frequency points as there are sampled points (N):}$$

$$\omega_k = \frac{2\pi k}{N\Delta t}, \quad k = 0, 1, \dots, N-1$$

Symmetry:

$$f_W\left(\frac{(N-k)2\pi}{T}\right) = \sum_{n=0}^{N-1} W(n\Delta t) F(n\Delta t) e^{-\frac{i(N-k)n2\pi}{N}} \cdot \Delta t = \sum_{n=0}^{N-1} W(n\Delta t) F(n\Delta t) e^{+\frac{ikn2\pi}{N}} \cdot \Delta t = f_W^*(k \cdot 2\pi/T)$$

The $(N-k)^{\text{th}}$ point in the spectrum is the same as the complex conjugate of the k^{th} point! Accordingly, it is redundant to evaluate all the N points in the frequency space, it is enough to take the $\omega_k = \frac{2\pi k}{N\Delta t}$, $k = 0, 1, \dots, \frac{N}{2}$ points. These are altogether $N/2+1$ points. Note, that the first ($k=0$) and the last ($k=N/2$) points from these are real, and the rest are complex. This means that we have N real sampled datapoints in the time-space, from which 2 real and $N/2-1$ complex spectral values are calculated. The latter has the equivalent information content of N real numbers, i.e. the "information content is preserved". This makes sense!

10

Discrete Fourier transformation (DFT)

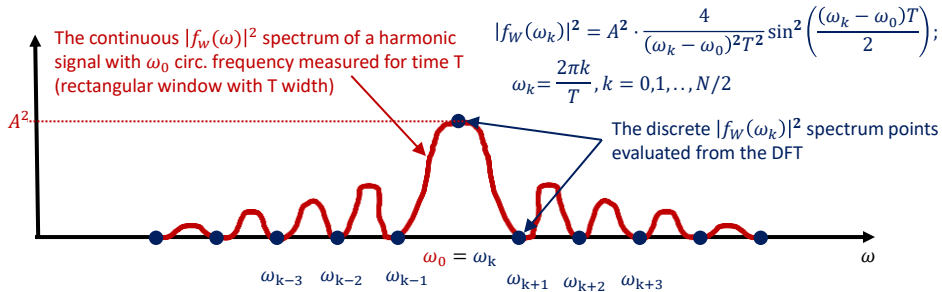
Revisiting the DFT formula:

$$f_W(\omega_k) \approx \sum_{n=0}^{N-1} W(n\Delta t) F(n\Delta t) e^{-i\omega_k n\Delta t} \cdot \Delta t,$$

$$\omega_k = \frac{2\pi k}{N\Delta t}, \quad k = 0, 1, \dots, \frac{N}{2}$$



$\omega_{\max} = 2\pi/(2\Delta t)$ Our DFT evaluation is in line with the sampling theorem.



In this example the discrete spectrum points are positioned at the zero values of $|f_W(\omega)|^2$ except for a single ω_k point, which is at the ω_0 circular frequency of the measured signal. In this case the value of the ω_k spectrum point gives A^2 , i.e. the amplitude of the signal can be directly read from the DFT. The $\omega_k = \omega_0$ condition yields:

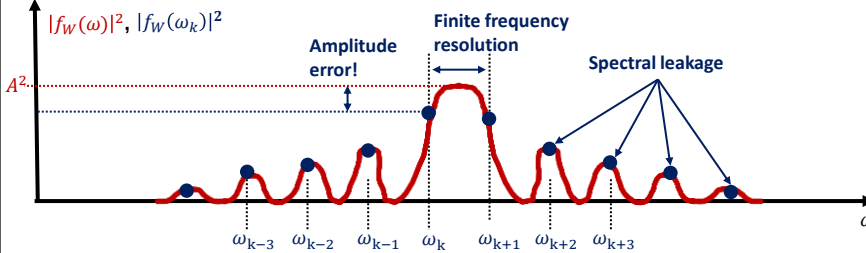
$$\omega_k = \frac{2\pi k}{T} = \omega_0 = \frac{2\pi}{T_0} \Rightarrow T = kT_0$$

This means, that the finite frequency resolution and the spectral leakage is not observed in the DFT, if the T measurement time is an integer multiple of the period of the signal.

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Discrete Fourier transformation (DFT)

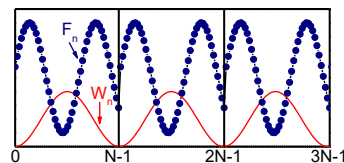
In a general case, we do not know the signal, so we cannot grant that the T measurement time is an integer multiple of the T_0 signal period. In this case the central DFT point is not at the peak of the real spectrum (ω_0), we cannot read the amplitude of the signal from the amplitude of the central DFT point, and the remote DFT points may catch the peaks of the spectral leakage.



A further property: the DFT of N datapoints is practically the same as the DFT of the periodic extension of the N datapoints:

$$\omega_k N\Delta t = 2\pi k \Rightarrow f_W(\omega_k) = \sum_{n=0}^{N-1} W(n\Delta t) F(n\Delta t) e^{-i\omega_k(n+N)\Delta t} \cdot \Delta t = \sum_{m=n+N}^{2N-1} W(m\Delta t - N\Delta t) F(m\Delta t - N\Delta t) e^{-i\omega_k m\Delta t} \cdot \Delta t$$

If the time T does not equal a multiple of the period of the measured signal, the periodic extension causes the boundaries to jump abruptly. This can also explain spectral leakage, i.e. the appearance of components other than the fundamental frequency. This is helped by the disappearing window function at the edge of the time window.



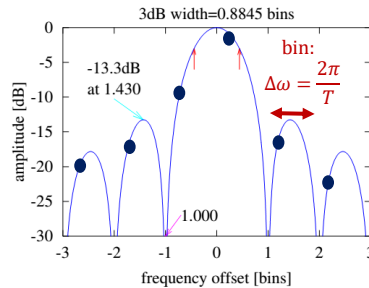
12

Window functions and their properties

Further on we analyze the properties of various window functions with similar figures, as the right one for the rectangular window. The DFT points are positioned equidistantly with one point / bin. Note the logarithmic vertical scale!

Which is the best window function? It depends on the goal of our analysis, do we want to have **good frequency resolution**, **small spectral leakage**, or **small amplitude error**?

Good frequency resolution is obtained if the central peak of $f_W(\omega)$ is as narrow as possible. This can be characterized by the distance between the 3dB suppression points, i.e. the frequencies where the power is halved ($10 \cdot \log(1/2) = -3.0102...$)



If we want **spectral leakage** to be **small**, we need a window function whose Fourier transform has the side peaks suppressed as much as possible. This can be characterized by the suppression of the first side peak ($-13.2\text{dB} \rightarrow 4.7\%$)

If we want to **accurately determine the amplitude of our signal** with ω_0 circ. frequency, we need to have a very flat $f_W(\omega)$ within the interval of $2\pi/T$ around the central frequency. The worst case is, when ω_0 is halfway between two discrete frequency points. Even in this case we can accurately read the signal amplitude if the window function is flat in the frequency space. In this case, however, our frequency resolution will be poor by definition.

Note: it is useful to normalize the window functions!

$$F(t) = A e^{i\omega_0 t} \Rightarrow f_W(\omega_0) = A \cdot w(0), \quad w(0) = \frac{T/2}{\int_{-T/2}^{T/2} W(t) dt}$$

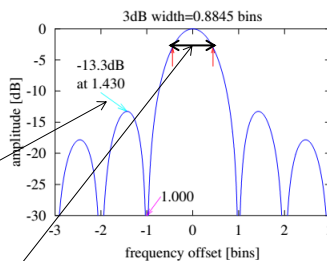
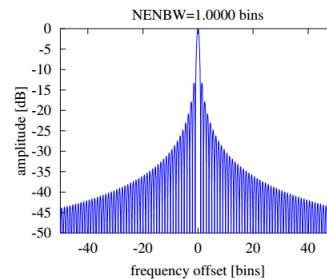
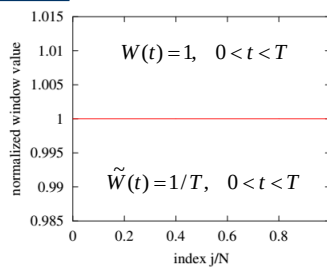
$$\text{Normalized window function: } \tilde{W}(t) = \frac{W(t)}{\int_{-T/2}^{T/2} W(t) dt}$$

Taking advantage of this, $f_{\tilde{W}}(\omega_0) = A$ is valid regardless of the choice of the window function, i.e. the signal amplitude can be easily read from the height of the central peak.

13

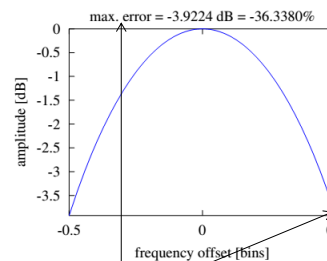
Rectangular window

Rectangular



The first side peak suppression: -13.3dB (4.7%). This is a **strong spectral leakage**.

Frequency resolution (-3dB point spacing): 0.8845 bin. This is the **best frequency resolution** for a certain T , the other window functions have worse frequency resolution!

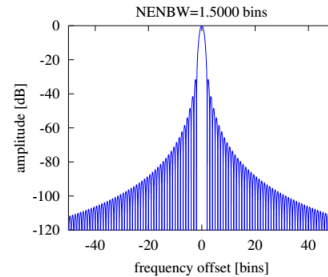
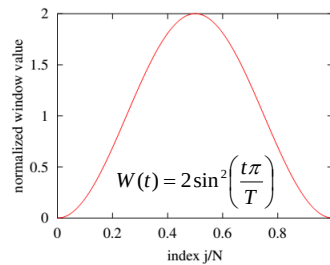


If ω_0 is halfway between two discrete frequencies, we measure 36% less amplitude than if ω_0 coincides with a discrete frequency. This is very **poor amplitude accuracy**.

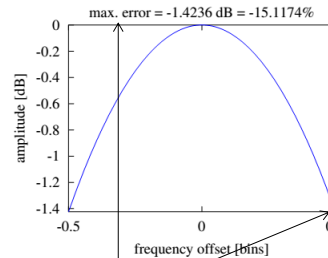
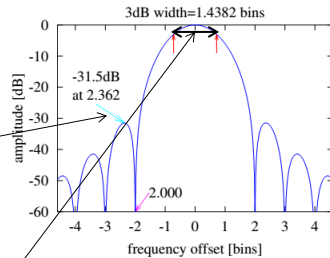
14

Hanning window

Hanning



The first side peak suppression: -31.5dB (0.07%). Relatively **low spectral leakage**, significantly **better than the rectangular window**.



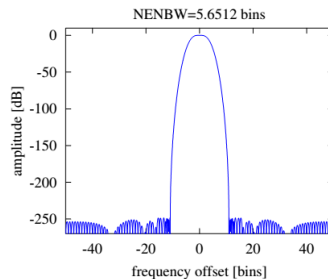
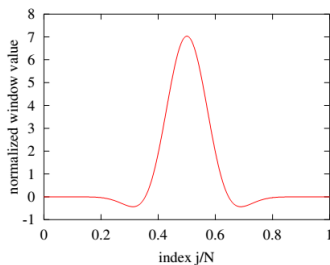
Frequency resolution (-3dB point spacing): 1.44 bin. This is worse than the rectangular window, but not too bad.

If ω_0 is halfway between two discrete frequencies, we measure 15% less amplitude than if ω_0 coincides with a discrete frequency. This is still a **poor amplitude accuracy**.

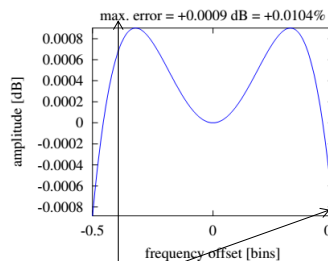
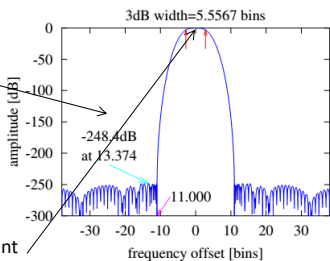
15

Flattop window

HFT248D



The first side peak suppression: -248dB. (1.6×10^{-25}) **Very low spectral leakage!**



Frequency resolution (-3dB point spacing): 5.5 bin. This is **very poor** frequency resolution! This is not a surprise, as the effective measurement time with nonzero window is just a fraction of T.

Wherever ω_0 is located between two discrete frequencies, the same value is obtained for the amplitude to within **0.01%**. The **flattop window is commonly used when good amplitude accuracy is the goal** and frequency resolution is less important!

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FFT (fast fourier transform)

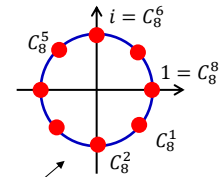
Reminder of the DFT:

$$f(k \cdot \Delta\omega) \approx \sum_{n=0}^{N-1} F(n\Delta t) e^{-ik\Delta\omega n\Delta t} \cdot \Delta t, \quad \Delta\omega = \frac{2\pi}{N\Delta t}$$

Δt is then dropped from the further calculations. (It can be done, if the amplitude is not important, just the spectral decomposition. Later, in noise measurements, it is not possible, as the noise is a physical quantity, and its true amplitude is important.)

$$f(k) = \sum_{n=0}^{N-1} F(n) e^{-i2\pi nk/N} = \sum_{n=0}^{N-1} F(n) (C_N)^{kn}, \quad k = 0 \dots N-1; N = 2^p; C_N = e^{-i2\pi/N}$$

The number of datapoints is a power of 2, we will see later, why.



These are complex numbers spaced equidistantly around the complex unit circle

Number of operations : 1 k point: N multiplication & N-1 addition, ~N „operations“. If we evaluate N k points: altogether ~N² operations. Now we ignore the previously discussed symmetry of the DFT, which would half the k points, as we will need all the N k points later for the FFT.

Trick: we calculate the DFT separately for the even (n=2r) and odd (n=2r+1) points:

$$f(k) = \sum_{r=0}^{N/2-1} F(2r) C_N^{2rk} + \sum_{r=0}^{N/2-1} F(2r+1) C_N^{(2r+1)k} = \sum_{r=0}^{N/2-1} F(2r) (C_N^2)^{kr} + C_N^k \sum_{r=0}^{N/2-1} F(2r+1) (C_N^2)^{kr}$$

$$f(k) = \underbrace{\sum_{r=0}^{N/2-1} F(2r) C_N^{kr}}_{f_{\text{even}}(k)} + C_N^k \underbrace{\sum_{r=0}^{N/2-1} F(2r+1) C_N^{kr}}_{f_{\text{odd}}(k)}$$

$$C_N^2 = e^{-i4\pi/N} = C_{N/2}$$

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FFT (fast fourier transform)

$f_{\text{even}}(k)$ and $f_{\text{odd}}(k)$ are each an N/2-point DFT, and it is enough to evaluate the k to 0..N/2-1, since e.g:

$$f_{\text{even}}(k - N/2) = \sum_{r=0}^{N/2-1} F(2r) C_{N/2}^{(k-N/2)r} = \sum_{r=0}^{N/2-1} F(2r) C_{N/2}^{kr} \underbrace{C_{N/2}^{-rN/2}}_1 = f_{\text{even}}(k) \quad \text{and similarly, for } f_{\text{odd}}(k)$$

That is, computing $f_{\text{even}}(k)$ and $f_{\text{odd}}(k)$ is (N/2)² + (N/2)² operations, and computing f(k) from the two is further N operations (see the C_N^k multiplication of f_{odd}), i.e. N²/2+N operations in total instead of the original N²!!

Since we have N=2^p data points, the above two N/2 point DFTs can be further divided into four N/4 point DFTs. Finally, the original N-point DFT can be reduced to N=2^p single-point DFTs. From these we can calculate f(k) with a small number of operations:

Elementary DFT number of points	To calculate the full DFT, i.e. f(k) number of operations:
N	N ²
$\frac{N}{2}$	$2\left(\frac{N}{2}\right)^2 + N = \frac{N^2}{2} + N$
$\frac{N}{4}$	$2\left(2\left(\frac{N}{4}\right)^2 + \frac{N}{2}\right) + N = \frac{N^2}{4} + 2N$
$\frac{N}{8}$	$2\left(2\left(2\left(\frac{N}{8}\right)^2 + \frac{N}{4}\right) + \frac{N}{2}\right) + N = \frac{N^2}{8} + 3N$
...	...
$\frac{N}{2^p} = 1$	$\frac{N^2}{2^p} + pN = \frac{N^2}{N} + N \log_2 N \approx N \log_2 N$

Example: N=10⁹ datapoints, assume that a single operation takes 1ns time:

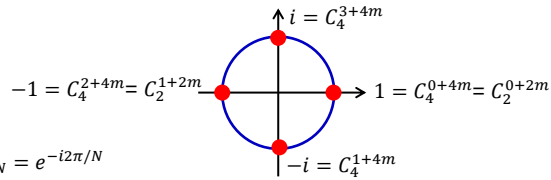
„brute force“ DFT calculation (N² operations):
10¹⁸ ns ~ 31 years

FFT algorithm: (N*log₂N operations):
30 * 10⁹ ns ~ 30 seconds

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Example: 4 point FFT

$$f_k = \sum_{n=0}^{N-1} F_n C_N^{kn}, k = 0..N-1; N = 2^p; C_N = e^{-i2\pi/N}$$



$$\begin{aligned} f_0 &= F_0 C_4^0 + F_1 C_4^0 + F_2 C_4^0 + F_3 C_4^0 = F_0 C_2^0 + F_2 C_2^0 + C_4^0 \cdot [F_1 C_2^0 + F_3 C_2^0] \\ f_1 &= F_0 C_4^1 + F_1 C_4^1 + F_2 C_4^1 + F_3 C_4^1 = F_0 C_2^1 + F_2 C_2^1 + C_4^1 \cdot [F_1 C_2^1 + F_3 C_2^1] \\ f_2 &= F_0 C_4^2 + F_1 C_4^2 + F_2 C_4^2 + F_3 C_4^2 = F_0 C_2^0 + F_2 C_2^0 + C_4^2 \cdot [F_1 C_2^0 + F_3 C_2^0] \\ f_3 &= F_0 C_4^3 + F_1 C_4^3 + F_2 C_4^3 + F_3 C_4^3 = F_0 C_2^1 + F_2 C_2^1 + C_4^3 \cdot [F_1 C_2^1 + F_3 C_2^1] \end{aligned}$$

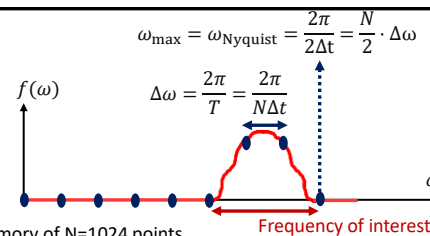
16 operations
 $f_{\text{even}}(k)$ 4 operations
 $f_{\text{odd}}(k)$ 4 operations

That is, the 16-operation 4-point DFT has been reduced to 12 operations by reducing the even and odd points back to a 2-2-point DFT (see FFT algorithm). However, if we are only interested in the last two frequency points (f_3, f_4), the direct count is 8 operations, but the FFT count saves only 2 operations (C_4^0 and C_4^1 multiplication), i.e. 10 operations are needed. This shows that the FFT algorithm is only effective if all frequency points from zero frequency to f_{max} are calculated. In other words, even if we want to evaluate only a limited frequency range of the spectrum, the FFT algorithm takes almost as many operations as if we evaluate the whole spectrum.

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FFT spectrum analyzer

The analyzed signal is sampled at Δt sampling intervals, and the spectrum is obtained by calculating the Fourier transform of the signal using the FFT algorithm. Here the maximum frequency in the spectrum is the Nyquist frequency, i.e. for high bandwidth a fast sampling is needed, whereas the good frequency resolution is granted by a long measurement time.



Example: we have an (old) spectrum analyzer with a total memory of $N=1024$ points

- If we sample for $T=1\text{ms}$ ($\Delta t=1\text{ms}/1024 \sim 1\mu\text{s}$ time resolution) we can resolve the FFT spectrum to components with frequency $f_k=k/T, k=0..N/2$, i.e. our frequency resolution is $\Delta f=1/1\text{ms}=1\text{kHz}$. With this resolution we can see the signal from DC to $f_{\text{max}}=512\text{kHz}$ Nyquist frequency.
- If we sample for $T=1\text{s}$ ($\Delta t=1\text{s}/1024 \sim 1\text{ms}$ time resolution) then in the FFT spectrum we have a frequency resolution of $\Delta f=1\text{Hz}$, with this resolution we see the signal from DC to $f_{\text{max}}=512\text{Hz}$.

That is: either we see the signal with good frequency resolution but only down to low frequency, or with poor frequency resolution but up to high frequency

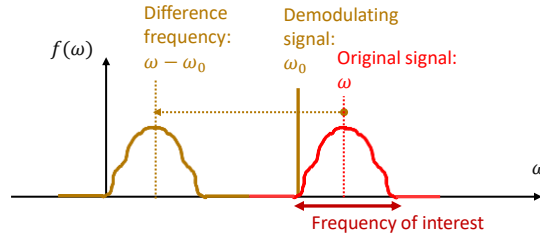
In the previous example, the limiting factor was the small memory of our (old) spectrum analyzer, because we could only record 513 discrete frequency points. Consider a new spectrum analyzer with a **memory of 1Msample** (10^6). With this, and a **sampling time of 1s**, we can see the signal from **DC to 0.5MHz** with a **resolution of 1Hz** in the frequency domain, i.e. we have good frequency resolution at high frequencies!

Problem: the FFT calculation is slow because of the many points. Moreover, when computing the FFT, we cannot compute only the **frequency components of interest** to us, since the FFT is only fast if we compute all the points at once, and **with fewer ω points the number of operations does not scale down with the number of points.**

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Heterodyne spectrum analysis

What if we want to see the signal not from DC but only for the frequency range of interest, but with good frequency resolution, and also avoiding the computation intensive FFT algorithm for millions of points?



Trick: Before calculating the Fourier transform, multiply the signal by a harmonic signal with circular frequency ω_0 .

The Fourier transform of the multiplication:

$$\tilde{f}_W(\omega) = \int_{-\infty}^{\infty} dt W(t) F(t) e^{i\omega_0 t} e^{-i\omega t} = f_W(\omega - \omega_0), \text{ where } f_W(\omega) = \int_{-\infty}^{\infty} dt W(t) F(t) e^{-i\omega t}$$

That is, the frequency component ω of the original signal appears in the Fourier transform of the multiplication shifted by ω_0 ! The frequency resolution is still determined by the sampling length T : $\Delta\omega \sim 2\pi/T$.

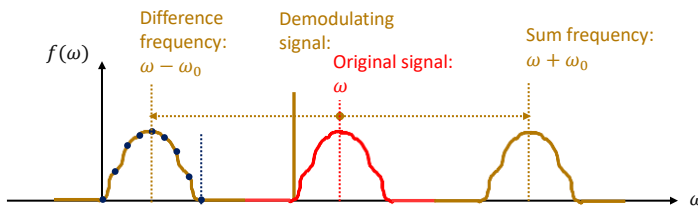
In measurement, we can only multiply by a real signal, so let us calculate the Fourier transform of a signal multiplied by a cosine signal:

$$\tilde{f}_W(\omega) = \int_{-\infty}^{\infty} dt W(t) F(t) \frac{\cos(\omega_0 t)}{e^{i\omega_0 t} + e^{-i\omega_0 t}} e^{-i\omega t} = \frac{1}{2} f_W(\omega - \omega_0) + \frac{1}{2} f_W(\omega + \omega_0)$$

That is, the difference and sum frequencies are also appearing. If the high-frequency component (sum frequency) in the multiplicative signal is suppressed by a low-pass filter and the FFT is applied to the resulting signal, then, e.g. with a measurement time of 1s, a memory of 1024 points and $f_0=500\text{kHz}$ "multiplication signal", the difference frequency signal $f_w(\omega - \omega_0)$ is displayed with a resolution of 1Hz from DC to 512Hz, i.e. the spectrum $f_w(\omega)$ of the original signal is recorded with a resolution of 1Hz in the range 500000-500512Hz! This is called **heterodyne detection**. As the signal of interest is shifted close to the DC domain, we can use slower sampling with less datapoints, which saves computation time, and we don't need fast A/D converters.

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Hybrid spectrum analyzer



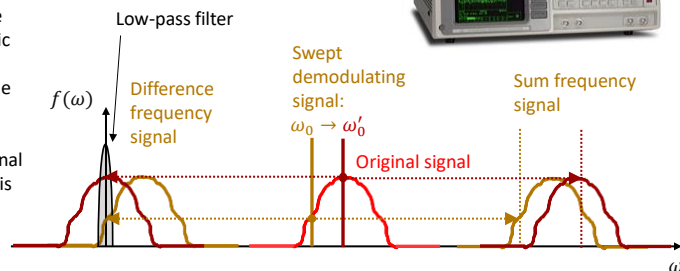
$$\Delta\omega = \frac{2\pi}{T} = \frac{2\pi}{N\Delta t}$$

$$\omega_{\max} = \frac{N}{2} \cdot \Delta\omega$$

The signal is down-converted with the fixed ω_0 frequency multiplying signal corresponding to the frequency window of interest, and the FFT algorithm is applied to the resulting difference frequency signal. Here, a slower sampling is enough to calculate the FFT of the difference frequency signal. The sum frequency signal and the leftovers of the original frequency signal in the product spectrum are suppressed by low-pass filters.

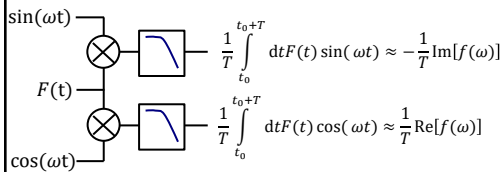
Swept-tuned spectrum analyzer

Using heterodyne detection, the signal is multiplied by a harmonic signal of frequency ω_0 , the DC component is measured from the multiplied signal by a low-pass filter (this gives the Fourier component of the measured signal of frequency ω_0). The spectrum is determined by tuning the frequency of the multiplying signal.



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Swept-tuned spectrum analyzer in more detail



The input signal of the spectrum analyzer ($F(t)$) is multiplied by a signal of circular frequency ω ($\cos(\omega t)$) and its 90 degree shift ($\sin(\omega t)$). The multiplied signals are passed through low-pass filters. A low-pass filter is equivalent to integrating the (multiplied) signal over a certain time T . It can be seen that the two outputs are proportional to the real and the imaginary parts of the Fourier transform of the signal $F(t)$ at frequency ω . By varying the circular frequency ω , we can map the full Fourier spectrum.

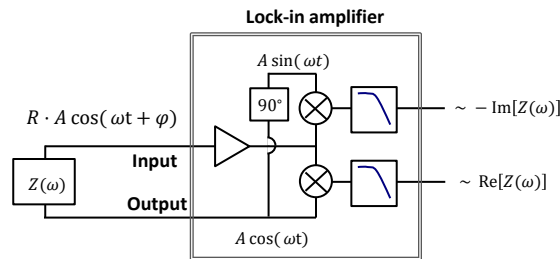
Lock-in amplifier

Practically the same thing is done by the so-called lock-in amplifier. The only difference is that $F(t)$ is not an arbitrary external signal, but a physical system is driven by the $\cos(\omega t)$ signal output by the lock-in and the response is measured at the input of the lock-in. For a linear system, the response function is a multiplication by a complex number ($Z(\omega)$), which can be described by an amplitude (R) and a phase shift (ϕ).

The lock in, after amplification, multiplies the signal at its input by the drive signal ($\sim \cos(\omega t)$) and its 90 degree shift ($\sim \sin(\omega t)$), passing the multiplied signals through a low-pass filter (for a digital lock in amplifier, this is effectively integration). The resulting signals are the real and the imaginary parts of the complex response function $Z(\omega)$. This method effectively filters out any interfering signal at a frequency other than that of the drive. This is called **phase-sensitive detection**.

$$\text{Drive: } \text{Re}[A \cdot e^{i\omega t}] = A \cdot \cos(\omega t)$$

$$\text{Response: } \text{Re}[Z(\omega) \cdot A \cdot e^{i\omega t}] = \text{Re}[Z(\omega)] \cdot A \cdot \cos(\omega t) - \text{Im}[Z(\omega)] \cdot A \cdot \sin(\omega t)$$



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Phase locked loop, PLL

Goal: to synchronize the output of the PLL to the input signal (this is also used in a lock-in, see next slide)

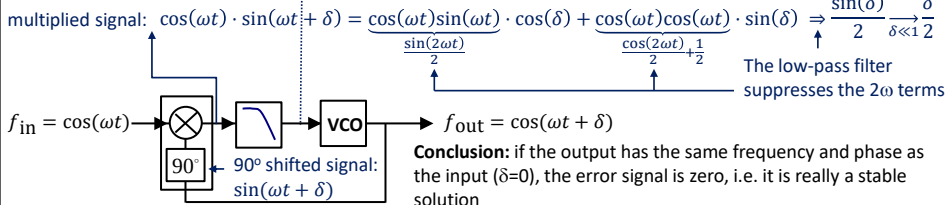
Phase detector: generates an error signal proportional to the phase difference between the output and input signal

Low-pass filter: filters the sum frequency from the multiplied signal

Voltage controlled oscillator: changes the output frequency with a speed proportional to the input voltage (i.e. the error signal), $df/dt \sim V_{in}$. This is done with a PI regulator (see temperature control in one of the first lectures) using a setpoint, where the error signal is zero.

Fixpoint: if the output and input signals are of the same frequency and phase, the phase detector gives a zero error signal, so the VCO does not change frequency. The output signal is then synchronized to the input signal. If the input signal is "noisy" around a given frequency, it will be "cleaned up" at the output.

Example:

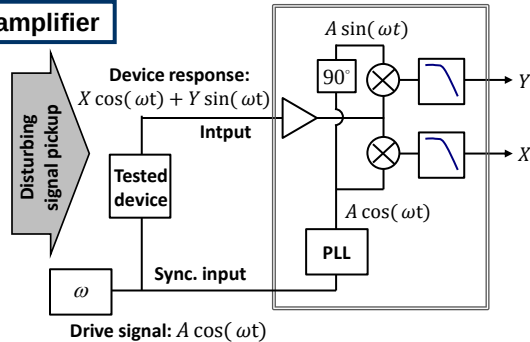


If, however, the output has the same frequency but different phase (nonzero constant δ), or moreover, the output is somewhat detuned in frequency (consider $\delta = \Delta\omega \cdot t + \delta_0$), the nonzero error signal is compensated by the PI loop of the PLL and the VCO will finally find the stable solution, where the output is "locked" to the input.

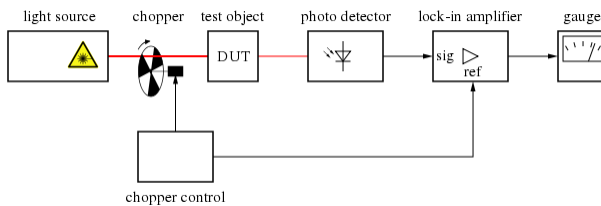
24

PLL synchronization in a lock-in amplifier

The lock in amplifier can also be synchronized to an external signal. We drive the system with an ω circular frequency signal (not necessarily harmonic). This signal is also connected to the synchronizing input of the lock-in synchronizer. The lock-in uses a PLL to synchronize its internal oscillator to the drive signal and performs the multiplication by its own synchronized (harmonic) signal, or by shifting it by 90 degrees.



Measuring a small signal in a noisy environment using a modulation technique: we illuminate an object with a laser and want to measure the transmitted light with a photodetector. However, the detector can also "see" light coming from elsewhere. We modulate our signal with a frequency f (e.g. we chop it up), the lock-in amplifier is synchronized to the modulating signal, so that the lock-in input of the detector only sees the effect of the modulated laser light, everything else is "filtered out".



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Higher harmonic measurement with a lock-in

Higher harmonic generation in a nonlinear system:

The response of a nonlinear system to harmonic driving also includes higher harmonics. With a lock-in amplifier and a $\cos(\omega t)$ drive, we can measure the $\cos(n\omega t)$ and $\sin(n\omega t)$ response, i.e. the amplitude and phase of the n^{th} harmonic.

Example: consider a circuit element with nonlinear current-voltage ($I(V)$) characteristics. To a DC voltage V_0 , we add a low amplitude AC signal:

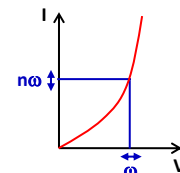
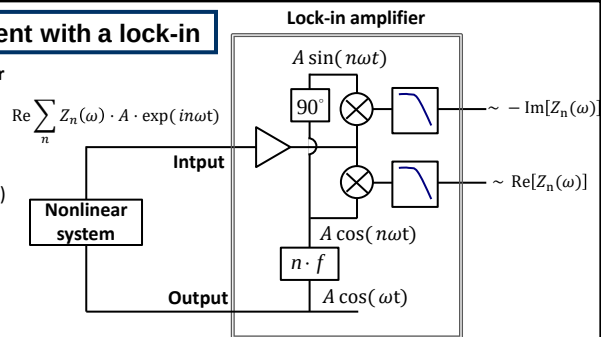
$$V(t) = V_0 + V_1 \cos(\omega t)$$

Measure the AC current with $n\omega$ circular frequency at the lock in input. What will be the amplitudes of the different harmonics?

$$I(V(t)) = I(V_0 + V_1 \cos(\omega t)) \approx I(V_0) + \left. \frac{dI}{dV} \right|_{V_0} \cdot V_1 \cos(\omega t) + \frac{1}{2} \left. \frac{d^2 I}{dV^2} \right|_{V_0} \cdot V_1^2 \cos^2(\omega t)$$

That is, the response at the fundamental frequency ($n=1$) will be proportional to the derivative of the $I(V)$ curve, and the second harmonic response ($n=2$) will be proportional to the second derivative of the $I(V)$ curve, the latter measuring the degree of nonlinearity.

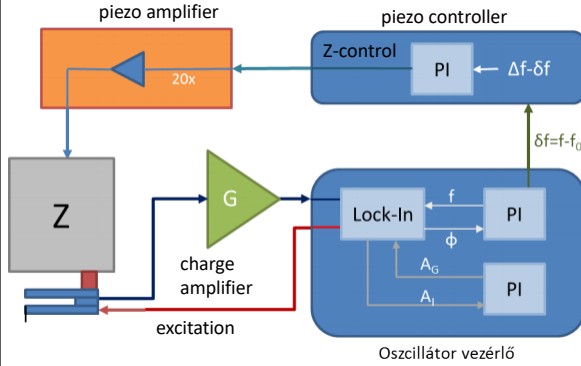
With this method, the first and second derivatives of the $I(V)$ curve can be measured directly while slowly varying V_0 , without numerically differentiating the $I(V)$ curve measured with a DC instrument.



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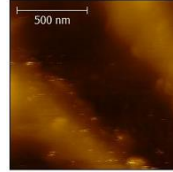
Atomic force microscope with a quartz tuning fork

The "clock signal" in the watches is provided by a tuning-fork-shaped quartz oscillator, which can be excited electrically due to its piezoelectric behaviour. By sticking an STM needle on one of the branches of the tuning fork, not only the tunnel current but also the force between the sample and the needle can be studied by the resonant frequency of the quartz oscillator. Based on the force measurement, we can even take images of insulating surfaces. To do this, a complex control scheme has to be implemented:



During scanning, the resonance frequency is varied due to the force between the needle and the sample. The PLL tells the actual resonance frequency (δf) relative to the free resonance frequency (f_0). The z-direction piezo is moved by means of a third PI controller so that the frequency detuning, i.e. the force between the needle and the sample, is constant. This corresponds to a Δf setpoint detuning. (Precisely: the frequency detuning is not sensitive to the force but to the spring constant, i.e. the force gradient)

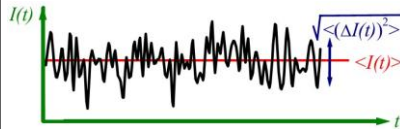
A PLL is used to excite the quartz oscillator continuously at its resonance frequency: i.e. a voltage with ω circular frequency and the current of the quartz oscillator is measured. A PI controller is used to control the frequency so that the phase shift is zero (i.e. it is excited at exactly the resonance frequency). Another PI controller is used to keep the amplitude of the oscillation constant.



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Spectral density of noise

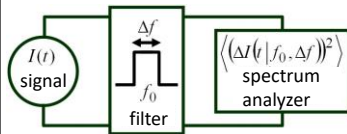
Usually, we are interested in the average value of a physical quantity (e.g. current), but around this average value we may experience fluctuations for various reasons:



The deviation of the current from the mean: $\Delta I = I(t) - \langle I(t) \rangle$

By definition: $\langle \Delta I(t) \rangle = 0$

Experimental definition of the spectral density of fluctuations: The signal $I(t)$ is passed through a band-pass filter with a narrow bandwidth Δf around a center frequency f_0 and the $\langle (\Delta I(t|f_0, \Delta f))^2 \rangle$ mean squared deviation of the such filtered signal is measured. The latter is proportional to the width of the frequency window Δf , and the proportionality factor is the $S_I(f_0)$ spectral density of the current noise at f_0 :



Spectral density of noise, or power spectrum density, PSD

Unit:

$$[S_I] = \frac{A^2}{Hz}$$

The mean squared deviation of the current over the full frequency range:

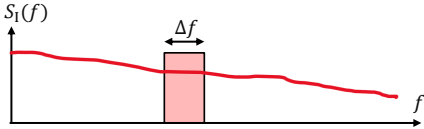
$$\langle (\Delta I(t))^2 \rangle = \int_0^\infty df S(f)$$

Similarly, for voltage noise:

$$\langle (\Delta V(t|f_0, \Delta f))^2 \rangle = S_V(f_0) \Delta f$$

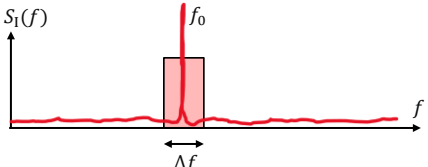
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Spectral density of noise



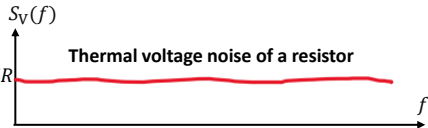
In the case of a peaked spectrum, i.e. a dominant frequency component at frequency f_0 , the total mean squared deviation within the band Δf does not depend on the width of the frequency window, only on whether f_0 is in that band or not. Therefore, a signal with a peaked spectrum is not noise, and does not even comply with the definition of noise.

Noise is a signal with continuous spectrum, with fluctuations of all frequencies, and the mean squared deviation of the various frequency components is additive. The continuous nature of the spectrum naturally yields that the overall mean squared deviation of the signal within a narrow Δf band is proportional to Δf , as included in the previous definition of the spectral density:

$$\langle (\Delta I(t|f_0, \Delta f))^2 \rangle = S_I(f_0) \Delta f$$


Thermal noise (will come later...)





If we take a simple resistor with no signal applied to it, we expect to measure zero voltage and zero current across it. In contrast, due to thermal fluctuations, we see a finite current noise when we connect the two ends of the resistor with a current meter. The spectral density of this noise is $S_I = 4kTG$, where k is the Boltzmann constant, T is the absolute temperature and $G=1/R$ is the conductance. The noise density of this so-called thermal noise is frequency independent (white noise). Similarly, if we connect a voltmeter to the resistor, we experience voltage fluctuations. These have an $S_V = S_I \cdot R^2 = 4kTR$ spectral density due to the $\Delta V = \Delta I \cdot R$ relation (Ohm's law), and $S_V \sim (\Delta V)^2$; $S_I \sim (\Delta I)^2$ (noise definition).

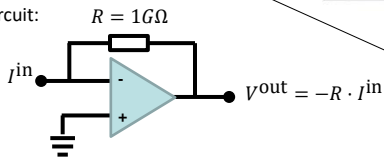
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Example: thermal noise of a current amplifier

FEMTO DLPCA-200 current amplifier used for scanning tunneling microscopy (STM) measurements.
 "10⁹" amplification (1GΩ feedback resistor),
 1nA input -> 1V output.
 Specifications:
 Input noise: 4.3fA/√Hz
 Bandwidth: 1kHz



Current amplifier circuit:



Reminder: the bandwidth is related to the RC time-constant of the feedback resistor and its stray capacitance

If the input current is zero, the output is the thermal voltage noise of resistor R:

$$\langle (\Delta V^{\text{out}})^2 \rangle = S_V \cdot \Delta f; S_V = 4kTR; \Delta V_{\text{rms}}^{\text{out}} = \sqrt{4kT \cdot R \cdot \Delta f}$$

The output would have the same noise if R had no noise, but the input current would be noisy. Normalized by the bandwidth, we get the corresponding effective input noise density:

$$\Delta I_{\text{rms}}^{\text{in}} = \Delta V_{\text{rms}}^{\text{out}} / R$$

$$S_I = \langle (\Delta I^{\text{in}})^2 \rangle / \Delta f$$

$$\sqrt{S_I} = \frac{\Delta I_{\text{rms}}^{\text{in}}}{\sqrt{\Delta f}} = \sqrt{\frac{4kT}{R}} \approx 4\text{fA}/\sqrt{\text{Hz}}$$

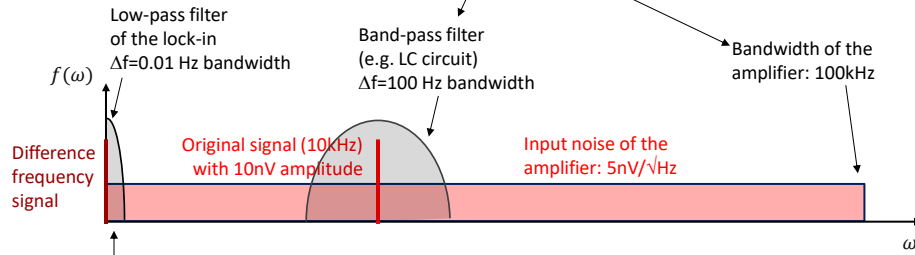
← This is the lowest noise you can get with this amplifier circuit at room temperature! For lower noise measurements, a low temperature amplifier is recommended.

30

Noise filtering considerations for a low noise amplifier – why should we use a lock-in?

Calculation example: we want to measure a signal with frequency 10kHz and amplitude 10nV with a low noise voltage amplifier with input noise $\sim 5\text{nV}/\sqrt{\text{Hz}}$ (see later for the definition of noise). The amplifier has a bandwidth of 100kHz and a gain of 1000 times. The amplified amplitude of our signal is $10\mu\text{V}$, while the amplifier's input noise results in a broadband noise output of 1.6mV rms ($5\text{nV}/\sqrt{\text{Hz}} \cdot \sqrt{100\text{kHz}} \cdot 1000$)!

A **bandpass filter** with a $Q=100$ quality factor (which is very good) can limit the bandwidth to $10\text{kHz}/Q=100\text{Hz}$. So, we see $50\mu\text{V rms}$ noise at the output of the amplifier ($5\text{nV}/\sqrt{\text{Hz}} \cdot \sqrt{100\text{Hz}} \cdot 1000$), which is still high compared to the $10\mu\text{V}$ amplified output of the true signal.



Using the **lock-in principle** (thanks to phase-sensitive detection), we can filter out the different frequency components much better than with the bandpass filter. The bandwidth is only limited by the integration time, if we measure on the scale of 100 seconds, a filtering bandwidth of 0.01 Hz is achieved, so we only see the amplifier output noise in this narrow band, which is $0.5\mu\text{V rms}$ ($5\text{nV}/\sqrt{\text{Hz}} \cdot \sqrt{0.01\text{Hz}} \cdot 1000$), a factor of 20 less than the signal to be measured.

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Current-current correlation function

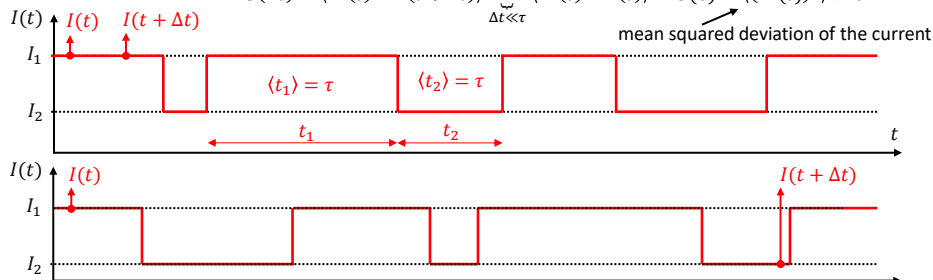
Current-current correlation function:

$$C(\Delta t) = \langle \Delta I(t) \cdot \Delta I(t + \Delta t) \rangle = \langle \Delta I(0) \cdot \Delta I(\Delta t) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T dt \Delta I(t) \cdot \Delta I(t + \Delta t), \quad \text{Unit: } [C(\Delta t)] = A^2$$

Example: random telegraph noise (RTN). The current randomly fluctuates between two values (I_1 and I_2), such, that the t_1 and t_2 times spent in the two states are random variables, but the mean value is t for both.

For $\Delta t \ll \tau$ there is a good chance, that $I(t + \Delta t) = I(t)$ (no switching), i.e.:

$$C(\Delta t) = \langle \Delta I(t) \cdot \Delta I(t + \Delta t) \rangle \approx \langle \Delta I(t) \cdot \Delta I(t) \rangle = C(0) = \langle (\Delta I(t))^2 \rangle > 0$$



For $\Delta t \gg \tau$ the $I(t + \Delta t)$ and the $I(t)$ values are completely independent from each other (like a random number generator for coin toss), i.e.:

$$C(\Delta t) = \langle \Delta I(t) \cdot \Delta I(t + \Delta t) \rangle \approx \underbrace{\langle \Delta I(t) \rangle}_0 \cdot \underbrace{\langle \Delta I(t + \Delta t) \rangle}_0 = 0$$

The current-current correlation function is the mean squared deviation of the current for small Δt , it decays to zero for $\Delta t \rightarrow \infty$, and the transition between the two happens around the characteristic time, so-called correlation time of the system, τ in this case. This is frequently described by an exponential correlation function:

$$C(\Delta t) = \langle \Delta I^2 \rangle \cdot e^{-\frac{|\Delta t|}{\tau_c}}$$

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Properties of the current-current correlation function

$C(0) = \langle (\Delta I(t))^2 \rangle > 0$, Mean squared deviation of the current

$$|C(\Delta t)| < C(0), C(\Delta t) = C(-\Delta t), \lim_{\Delta t \rightarrow \infty} C(\Delta t) = 0$$

The Fourier transform of the correlation function:

$t' = t$ variable change

Unit:

$$c(\omega) = \int_{-\infty}^{\infty} dt e^{-i\omega t} C(t) = \int_0^{\infty} dt e^{-i\omega t} C(t) + \int_0^{\infty} dt' e^{i\omega t'} C(t') = 2 \int_0^{\infty} dt C(t) \cos(\omega t) \quad [c(\omega)] = \frac{A^2}{Hz}$$

$c(\omega)$ is real, $c(\omega) = c(-\omega)$ from the cosine formula these properties follow

Inverse transformation: $C(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega e^{i\omega t} c(\omega)$

From this the mean squared deviation of the current: $\langle (\Delta I(t))^2 \rangle = C(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega c(\omega) = \frac{1}{\pi} \int_0^{\infty} d\omega c(\omega)$

On the other hand, the definition of the spectral density of noise yielded the

$$\langle (\Delta I(t))^2 \rangle = \int_0^{\infty} df S_I(f) = \frac{1}{2\pi} \int_0^{\infty} d\omega S_I(\omega = 2\pi f) \text{ formula for the mean squared deviation of the current.}$$

Comparing the two, we realize that **the spectral density of the current noise is twice the Fourier transform of the current-current correlation function (a more rigorous proof is also available):**

$$S_I(\omega = 2\pi f) = 2c(\omega)$$

The factor two difference is due to the fact that in the experimental definition of S_I we integrate only for positive frequencies, while for $c(\omega)$ we integrate also for negative frequencies. The same considerations apply for the noise of any quantity, not just current.

33

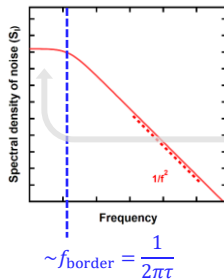
Examples of correlation functions

Exponentially decaying correlation function:

$$C(\Delta t) = C_0 \cdot e^{-\frac{|\Delta t|}{\tau}} = \langle \Delta I^2 \rangle \cdot e^{-\frac{|\Delta t|}{\tau}}$$

$$S(\omega) = 2c(\omega) = 2 \int_0^{\infty} dt e^{-i\omega t} C_0 e^{-\frac{t}{\tau}} + 2 \int_0^{\infty} dt e^{i\omega t} C_0 e^{-\frac{t}{\tau}} = \dots = \frac{4C_0\tau}{1 + \omega^2\tau^2}$$

$$\text{For } \omega \ll 1/\tau: S_0 = 4C_0\tau$$

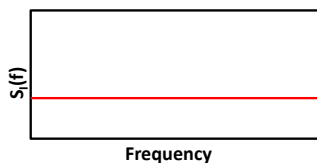


This is the so-called **Lorentzian noise spectrum**. Below the border frequency ($\sim$ inverse correlation time) a frequency independent noise is observed, here it looks "white noise". At higher frequencies the noise power decays with $1/f^2$.

White noise: Dirac-delta correlation function, i.e. current noise values at different times are completely uncorrelated ("zero correlation time"):

$$C(\Delta t) = C_0 \delta(\Delta t)$$

$$S(\omega) = 2c(\omega) = 2 \int_{-\infty}^{\infty} dt e^{-i\omega t} C_0 \delta(t) = 2C_0 = S_0$$



White noise: the spectrum is constant (frequency independent) in the entire frequency range

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How to calculate the noise spectrum in practice by FFT of the measured signal?

Fourier transform of the current deviation: $\Delta i(\omega) = \int_{-\infty}^{\infty} dt \Delta I(t) e^{-i\omega t}$

And the expected value of its absolute value squared for a number of independent measurements:

$$\begin{aligned} \langle |\Delta i(\omega)|^2 \rangle &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dt' dt \frac{\langle \Delta I(t) \Delta I(t') \rangle}{c(t'-t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega' c(\omega') \exp(i\omega'(t'-t))} e^{-i\omega t} e^{i\omega t'} = \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dt d\omega' c(\omega') e^{-i(\omega+\omega')t} \underbrace{\int_{-\infty}^{\infty} dt' \frac{e^{i(\omega+\omega')t'}}{2\pi}}_{\delta(\omega+\omega')} = c(\omega) \int_{-\infty}^{\infty} dt \end{aligned}$$

$$\langle |\Delta i(\omega)|^2 \rangle = c(\omega) \cdot \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} dt \quad S_I(\omega) = 2c(\omega) = \lim_{T \rightarrow \infty} \frac{2}{T} \left| \int_{-T/2}^{T/2} dt \Delta I(t) e^{-i\omega t} \right|^2$$

That is, $S_I(\omega)$ is proportional to the expected value of the Fourier transform's absolute value squared, but for an infinite-time Fourier integral the proportionality factor diverges.

What about approximating the above formula for $S_I(\omega)$ by the DFT (or FFT) of the signal as follows? We will check the proper normalization on the next slide.

$$PSD = S_I(\omega) \approx \frac{2}{N\Delta t} \left| \sum_{n=0}^{N-1} F(n\Delta t) e^{-i\omega n\Delta t} \Delta t \right|^2 = \frac{2\Delta t}{N} \left| \sum_{n=0}^{N-1} F(n\Delta t) e^{-i\omega n\Delta t} \right|^2$$

This is the practical method of calculating the noise from the data in a measurement

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Normalization of the DFT-based noise formula for various window functions

Calculating the spectral density of noise for a general window function:

$F(t) = N(t) \leftarrow$ Our signal is white noise (frequency independent noise density, time uncorrelated current fluctuation) Of course, it is sufficient to satisfy the white noise condition at the bandwidth corresponding to the sampling rate)

Fourier transform of the signal multiplied by the window function: $n_W(\omega) = \int_{-\infty}^{\infty} W(t) N(t) e^{-i\omega t} dt$

$$\langle |n_W(\omega)|^2 \rangle = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dt dt' W(t) W(t') \frac{\langle N(t) N(t') \rangle}{c(t'-t) = S_0 \delta(t'-t)/2} e^{-i\omega t} e^{i\omega t'} = \frac{S_0}{2} \int_{-\infty}^{\infty} W^2(t) dt$$

For white noise

This formula is already decent, as it includes the finite measurement time through the window function, and therefore it can be definitely approximated by the discrete sum of the DFT:

$$PSD = S_0 = \frac{2 \langle |n_W(\omega_0)|^2 \rangle}{\int_{-\infty}^{\infty} W^2(t) dt} \approx \frac{2}{\sum_{n=0}^{N-1} W^2(n\Delta t)} \left| \sum_{n=0}^{N-1} W(n\Delta t) F(n\Delta t) e^{-i\omega n\Delta t} \Delta t \right|^2$$

For a normalized square window $W(t) = 1/T$ is used during the time of the measurement, $\int_{-\infty}^{\infty} W^2(t) dt = 1/T$, but $W(t)$ and Δt also appear in the sum. Collecting all these to the prefactor, really the formula of the previous slide follows:

$$PSD = S_I(\omega) \approx \frac{2\Delta t}{N} \left| \sum_{n=0}^{N-1} F(n\Delta t) e^{-i\omega n\Delta t} \right|^2$$

But this formula is invalid if the DFT is calculated by another window. For a Hanning window the normalization differs by a factor of 1.5, please check if you wish!

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Detour: Drude model

Calculating R resistance from σ conductivity (L and A are the length and cross-section of the sample):

$$R = G^{-1} = \frac{1}{\sigma} \frac{L}{A}$$

Ohm's law for current density: $\mathbf{j} = \sigma \mathbf{E}$

A simple microscopic model of electrical conduction (Drude model):

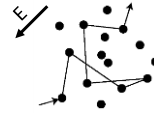
In a zero electric field, the average momentum of electrons is zero. In finite field, electrons gain momentum in the direction of the field strength (drift momentum). However, as they move through the crystal lattice, they undergo collisions on average at intervals τ , after which their momentum points in random directions (e.g. scattering by lattice defects, impurity atoms, lattice vibrations, etc.), i.e. the average drift momentum gained in the electric field between two collisions is lost by the electrons in time τ due to the collisions:

Average drift velocity gained between two collisions

$$\mathbf{p}_{\text{drift}} = m \mathbf{v}_{\text{drift}} = -e \mathbf{E} \tau \Rightarrow \mathbf{v}_{\text{drift}} = -\frac{e \tau}{m} \mathbf{E}$$

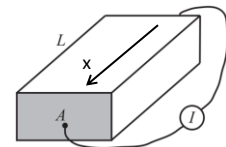
Key parameter: τ . The more collisions occur per unit time (i.e. the smaller τ), the greater the resistance.

$$\mathbf{j} = -ne \mathbf{v}_{\text{drift}} = \frac{ne^2 \tau}{m} \mathbf{E} \Rightarrow \sigma = \frac{ne^2 \tau}{m} \leftarrow \text{conductivity in the Drude picture}$$



The basic concept of thermal noise

In thermal equilibrium, without an external field, the average momentum of electrons is zero, but due to interaction with a finite temperature crystal lattice, electrons can gain random momentum, which is lost in time τ due to collisions. Thus, at a "snapshot", the total momentum of electrons moving in a given direction may be greater than that of electrons moving in the opposite direction. The momentum of the individual electrons on a snapshot is preserved on the time-scale of τ and lost afterwards. For this reason, in the arrangement shown in the figure (the two ends of a metal sample of length L with cross-section A are short-circuited), we will observe a random current fluctuation with a time average of zero.



Again, τ is the key parameter, the correlations in the current are lost on the characteristic time-scale of τ .

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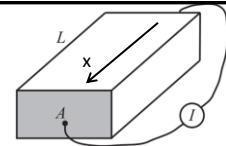
Calculation of thermal noise

Current density of a single electron in the x direction:

$$j_x(t) = -nev_x(t) = -\frac{1}{LA} ev_x(t);$$

Current of a single electron:

$$I_1(t) = A j_x(t) = -\frac{ev_x(t)}{L}$$



The current-current correlation function for a single electron:

$$C_1(\Delta t) = \langle I_1(t) I_1(t + \Delta t) \rangle = \frac{e^2}{L^2} \langle v_x(t) v_x(t + \Delta t) \rangle \approx \frac{e^2}{L^2} \langle v_x^2(0) \rangle \exp(-\Delta t / \tau)$$

The collisions cause the correlation to be lost in characteristic time τ

The correlation function for N electrons, assuming that the electrons are independent of each other:

$$C(\Delta t) = N C_1(\Delta t) = \frac{Ne^2}{L^2} \langle v_x^2(0) \rangle \exp(-\Delta t / \tau) = \frac{N}{AL} \frac{e^2 \tau A}{m L \tau} \underbrace{m \langle v_x^2(0) \rangle}_{kT} \exp(-\Delta t / \tau) = G \frac{kT}{\tau} \exp(-\Delta t / \tau)$$

Previously we calculated the noise density for an exponentially decaying correlation function:

$$S_1(\omega) = \frac{4kTG}{1 + \omega^2 \tau^2} \approx_{\omega \ll 1/\tau} 4kTG$$

Later, we will see from solid state physics that we have miscalculated at two points:

$$m \langle v_x^2(0) \rangle \neq kT, \text{ rather: } m \langle v_x^2(0) \rangle \approx \frac{mv_F^2}{3} \quad \frac{mv_F^2}{2} = E_F$$

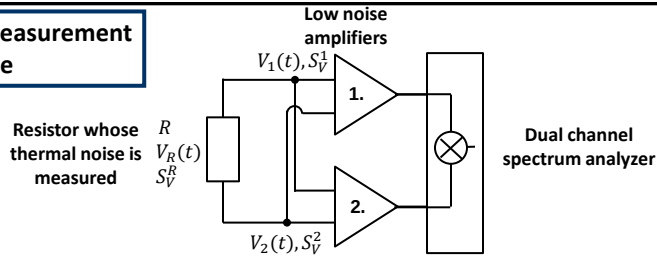
Fermi velocity Fermi energy

In addition, only about $N^* kT / E_F$ of the total N electrons can be considered independent, the rest are "frozen" and contribute zero to the current. These two errors roughly compensate each other.

Despite the miscalculation, we have learned an important conclusion: the correlation time of noise is the same τ time between the collisions, which is the key parameter of the G conductance as well. So, it is understandable that thermal noise depends not only on temperature but also on conductance.

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Cross-correlation measurement technique



The resistor's voltage fluctuates over time due to thermal noise: $V_R(t)$

The input noise of the amplifiers ($V_1(t)$ és $V_2(t)$) is added to the thermal noise of the resistor. The input noise of the two amplifiers is independent of each other! Like this, the two amplifiers output $A(V_R(t) + V_1(t))$ and $A(V_R(t) + V_2(t))$ voltage, respectively.

The dual-channel spectrum analyzer calculates a cross-correlation, i.e. it examines the spectrum of the product of two signals! As an illustration of the the method, let's just look at the expected value of the cross-product:

$$\langle A(V_R(t) + V_1(t)) \cdot A(V_R(t) + V_2(t)) \rangle = A^2 \underbrace{\langle V_R^2(t) \rangle}_{S_V^R \cdot BW} + \underbrace{A^2 \langle V_R(t) \rangle \langle V_2(t) \rangle + A^2 \langle V_1(t) \rangle \langle V_R(t) \rangle + A^2 \langle V_1(t) \rangle \langle V_2(t) \rangle}_0$$

If only one amplifier were used, the output voltage would have the mean squared deviation:

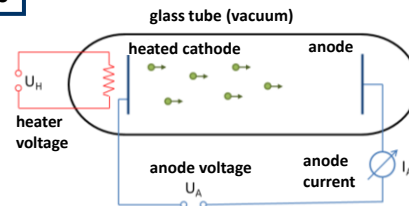
$$\langle A^2 (V_R(t) + V_1(t))^2 \rangle = A^2 \underbrace{\langle V_R^2(t) \rangle}_{S_V^R \cdot BW} + A^2 \underbrace{\langle V_1^2(t) \rangle}_{S_V^1 \cdot BW}$$

That is, for a single amplifier, the sum of the noise density of the sample and the input noise density of the amplifier is measured, while for a cross-correlation of two amplifiers, the noise of the amplifiers is cancelled out. When measuring noise, the amplifier input noise is usually comparable to or greater than the noise under test.

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Shot noise of independently emitted electrons

The shot noise is related to the quantized electron charge, which results in current of individual "electron shots" instead of a constant noiseless current. Schottky made the first shot noise measurement in 1918 measuring the current noise of a vacuum tube. Electrons emitted randomly from a heated cathode are driven by the applied voltage to the anode, where a current is detected. In this case the electrons are emitted randomly and independently of each other, so the electron incidences are well described by the Poisson distribution:



$$P_N(t) = \frac{t^N}{\tau^N N!} e^{-t/\tau}$$

← The probability that N electrons are emitted within time t, such that the average time between the independent electron emissions is τ

This formula can be proved by induction

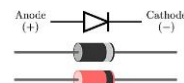
$$\left[P_N(t + dt) = P_{N-1}(t) \frac{dt}{\tau} + P_N(t) \left(1 - \frac{dt}{\tau} \right) \Rightarrow \frac{dP_N(t)}{dt} = \frac{P_{N-1}(t) - P_N(t)}{\tau}, \dots \right]$$

A special property of the Poisson distribution: the mean value of N for a certain time is the same as its mean squared deviation:

$$\langle N \rangle = \sum N \cdot P_N(t) = \frac{t}{\tau}, \quad \langle (\Delta N)^2 \rangle = \langle N^2 \rangle - \langle N \rangle^2 = \langle N \rangle$$

Note, that $\langle (\Delta N)^2 \rangle$ is proportional to the current noise, whereas $\langle N \rangle$ is proportional to the mean current, i.e. the $S_I \sim I$ relation is expected for the shot noise of independently emitted electrons ("Poisson noise")

Independent electron emissions are not only available in a vacuum tube, but also in other systems, like a semiconductor diode at small enough currents.



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Detour: signal filtering by convolution with a filtering window

What happens to our signal when we smooth it, i.e. average the adjacent points together?

Mathematically, we convolve our signal with a function $g(t)$, which is a Δt wide filtering window:

$$I_{\text{out}}(t) = \int_{-\infty}^{\infty} I_{\text{in}}(t') g(t-t') dt', \quad g(t) = \begin{cases} \frac{1}{\Delta t}, & \text{if } |t| \leq \frac{\Delta t}{2} \\ 0, & \text{otherwise} \end{cases} \Rightarrow I_{\text{out}}(t) = \frac{1}{\Delta t} \int_{t-\Delta t/2}^{t+\Delta t/2} I_{\text{in}}(t') dt'$$

We have seen earlier that the Fourier transform of a product is equal to the convolution of the Fourier transforms.

It can be shown that the Fourier transform of the convolution is the product of the Fourier transforms:

$i_{\text{out}}(\omega) = i_{\text{in}}(\omega) g(\omega)$
 ↑ ↑
 FT of the FT of the FT of the
 filtered original filtering
 signal signal window

The noise density is proportional to the absolute value squared of the currents Fourier transform:

$$S_{\text{out}}(\omega) = S_{\text{in}}(\omega) |g(\omega)|^2$$

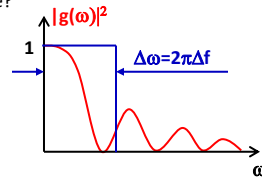
Earlier, we have seen: $g(\omega) = \int_{-\infty}^{\infty} dt g(t) e^{-i\omega t} = \frac{1}{\Delta t} \int_{-\Delta t/2}^{\Delta t/2} dt e^{-i\omega t} = \frac{2}{\omega \Delta t} \sin\left(\frac{\omega \Delta t}{2}\right)$

How much noise do we see at the output of the filter over the full frequency range?

For the same amount of output noise, what rectangular filter of width $\Delta\omega = 2\pi\Delta f$

would be needed assuming white input noise with S_0 density? This is the so-called **equivalent noise bandwidth, ENBW**.

$$\langle (\Delta I(t))^2 \rangle = \frac{1}{2\pi} \int_0^{\infty} d\omega S_{\text{out}}(\omega = 2\pi f) = \frac{S_0}{\pi \Delta t} \int_0^{\infty} dx \frac{\sin^2(x)}{x^2} = \frac{S_0}{2\Delta t} = S_0 \Delta f$$



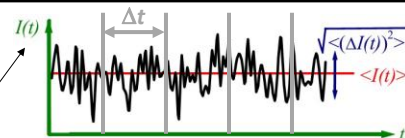
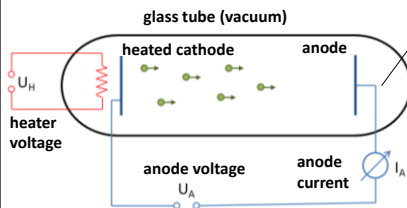
That is, the **equivalent noise bandwidth** of the filter:

$$ENBW = \Delta f = \frac{1}{2\Delta t}$$

Similar to the Nyquist-Shannon sampling theorem: at Δt sampling the signal is resolved up to $f_{\text{max}} = 1/2\Delta t$. But here we are not sampling at Δt intervals, rather we smooth the signal with a Δt filtering window. This acts as an antialiasing filter suppressing the spectrum above $f_{\text{max}} = 1/2\Delta t$.

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Independent emission of electrons: Poisson noise



We sample the current of the vacuum tube with time resolution Δt , measuring only the integrated current in a sampling interval:

$$I = Ne/\Delta t$$

Where N is described by the Poisson distribution:

$$P_N(\Delta t) = \frac{(\Delta t)^N}{\tau^N N!} e^{-\Delta t/\tau}$$

The average current over many sampling intervals Δt : $\langle I \rangle = \langle N \rangle e/\Delta t = e/\tau$

And the measured mean squared deviation is: $\langle (\Delta I)^2 \rangle = \langle (\Delta N)^2 \rangle e^2 / (\Delta t)^2 = \langle N \rangle e^2 / (\Delta t)^2 = \langle I \rangle e / \Delta t$

According to the definition of noise density:

$$\langle I \rangle e / \Delta t = \langle (\Delta I)^2 \rangle = \int_0^{f_{\text{max}}=1/2\Delta t} S_I(f) df$$

If we measure using Δt interval averaging, this is our bandwidth

This should be true for any Δt . This is only possible if:

$$S_I = 2e\langle I \rangle$$

This formula provides a unique possibility to determine the "e" electron charge from the measurement of the Poisson noise:

- This measurement was first carried out by Schottky in vacuum diodes (1918).
- You can try this measurement using a semiconductor diode in the "Measurement techniques laboratory course" together with the determination of the "k" Boltzmann constant by thermal noise measurement.

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Resistance fluctuations, 1/f-type noise

In a field-biased transistor, the gate dielectric may have charge traps present. An electron can be trapped in or released from a charge trap with a characteristic time τ , causing a resistance fluctuation with a correlation function of characteristic time τ :

$$C_R(\Delta t) = \langle \Delta R(t) \cdot \Delta R(t + \Delta t) \rangle = \langle \Delta R^2(t) \rangle \cdot \exp(-\Delta t/\tau)$$

From this the density of the resistance

noise is obtained by Fourier

transformation (Lorentzian noise spectra):

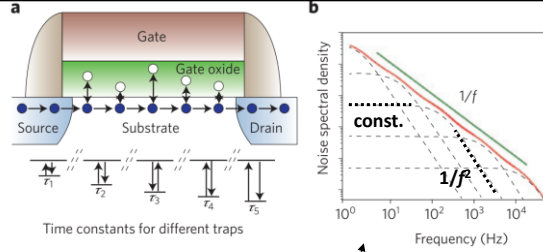
$$S_R(\omega) = 2C_R(\omega) = \frac{4 \langle \Delta R^2(t) \rangle \tau}{1 + \omega^2 \tau^2}$$

Adding together the Lorentzian noise spectra of many different processes with different τ , we get a summed spectrum with shallower frequency dependence than $1/f^2$, the so-called 1/f-type noise. This 1/f-type noise is present in many systems, even for completely different physical reasons, and often causes serious difficulties in measurements. For example, before the introduction of GaN/AlGaN high bandwidth semiconductor devices in communication technologies, years of research and development were needed to reduce the 1/f noise of these devices by about 5 orders of magnitude. Note, that it is not necessarily a truly $1/f$ frequency dependence, rather $1/f^\gamma$ with γ being close to unity.

If voltage noise is measured at constant current, then $\Delta V = I \cdot \Delta R$:

$$S_V(f_0) = \langle (\Delta V(t | f_0, \Delta f))^2 \rangle / \Delta f = I^2 \cdot \langle (\Delta R(t | f_0, \Delta f))^2 \rangle / \Delta f = I^2 \cdot S_R$$

That is, since 1/f noise is basically resistance noise due to resistance fluctuations, the related voltage noise scales by the square of the driving current, according to Ohm's law!



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Aliasing problem in noise measurements, antialiasing filter

The DFT evaluates the spectrum within the 0 and $\omega_{\max} = 2\pi/(2\Delta t)$ interval at discrete, $2\pi k/(N\Delta t)$ circular frequencies. What happens, if components with circular frequency higher than $\omega_{\max}$ are also present in the signal?

Let us consider a pure harmonic signal:

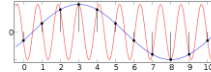
$$F(t) = A_0 e^{i\omega_0 t}, \quad f(\omega) = A_0 2\pi \delta(\omega - \omega_0)$$

Now let's see how the spectrum changes in practice when calculating the DFT. It is important to note that a data acquisition card or oscilloscope basically samples at Δt intervals for $\ll \Delta t$ times, and it is not the case that it averages the signal over Δt times!

$$DFT(\omega_k) = \sum_{n=0}^{N-1} A_0 e^{i\omega_0 n \Delta t} W(n \Delta t) e^{-i\omega_k n \Delta t} \Delta t \quad \text{notice, that: } DFT(\omega_k) = DFT\left(\omega_k + \frac{2\pi m}{\Delta t}\right)$$

where m is an integer, and furthermore: $DFT(\omega_k) = DFT^*(-\omega_k) \rightarrow |DFT(\omega_k)| = |DFT(-\omega_k)|$

Thus, it can be seen that any signal with a high circular frequency is seen as if it were a signal in the $[0, \omega_{\max} = \frac{2\pi}{2\Delta t}]$ domain with its own amplitude A_0 . This is the phenomenon of aliasing.



The components in the $(2m) \frac{2\pi}{2\Delta t} < \omega < (2m+1) \frac{2\pi}{2\Delta t}$, $m = 1, 2, \dots$ intervals appear at the $\omega_m = \omega - \frac{2\pi m}{\Delta t}$

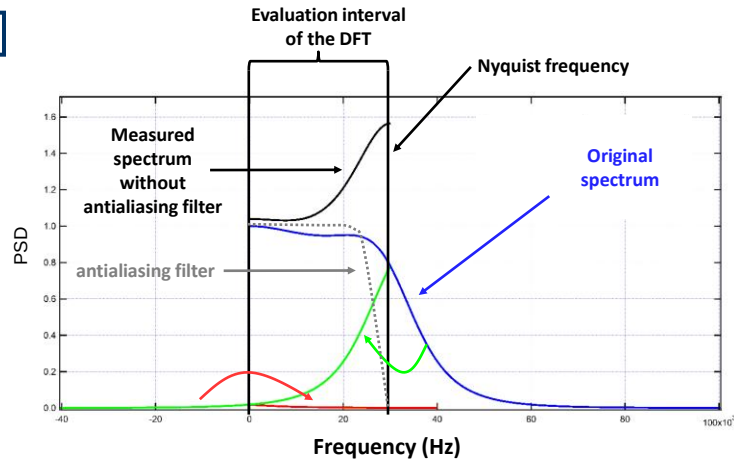
circular frequencies, while the $(2m-1) \frac{2\pi}{2\Delta t} < \omega < (2m) \frac{2\pi}{2\Delta t}$, $m = 1, 2, \dots$

interval converts to $\omega_m = \frac{2\pi m}{\Delta t} - \omega$ circular frequencies.

When measuring noise, we see a continuous spectrum, and each frequency component is in independent phase, so the power density of the frequency components converted from high frequency due to aliasing, i.e. the square of their Fourier transformed absolute value, is added to the real noise density to be measured at a given frequency!!!

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Antialiasing filter



If the measured noise also contains components above the Nyquist frequency ($f_{\max} = 1/2\Delta t$), the aliasing phenomenon causes the components above $f_{\max}$ to be “folded” into the frequency range of the DFT causing a completely false signal. Solution: the signal should be measured with an anti-aliasing filter which cuts off right at the Nyquist frequency, removing the higher frequency components, which would cause aliasing.